

PRAKTIKUM SISTEM OPERASI

MODUL 4

(MEMBACA SOURCE CODE XINU)



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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

DIREKTORAT KAMPUS PURWOKERTO

UNIVERSITAS TELKOM

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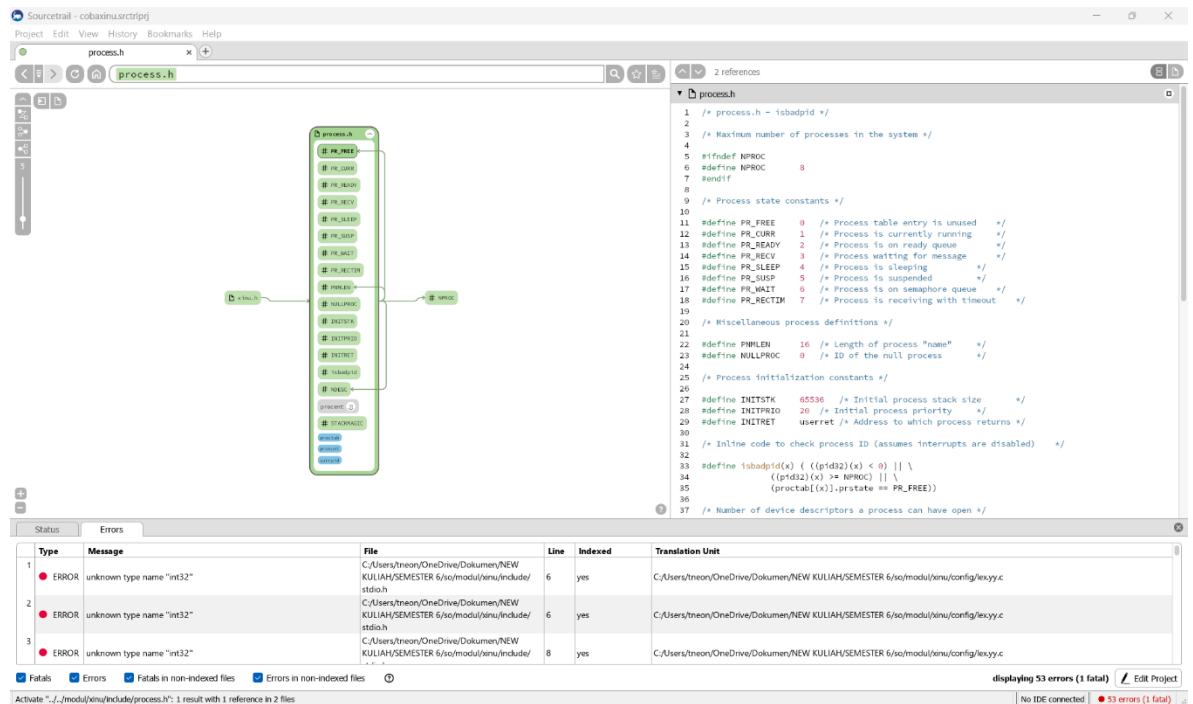
JURNAL 4

1. Apa nama image yang dihasilkan setelah melakukan kompilasi pada Xinu? Berapa ukuran file tersebut?

```
xinu@xinu-develop-end:~/xinu/compile$ ls
bin binaries ld.script Makefile version xinu.elf xinu.map
xinu@xinu-develop-end:~/xinu/compile$ ls -lh
total 340K
drwxr-xr-x 2 xinu xinu 4.0K May  8 2025 bin
drwxr-xr-x 2 xinu xinu 12K Jun  1 03:32 binaries
-rw-r--r-- 1 xinu xinu 1.6K May  8 2025 ld.script
-rw-r--r-- 1 xinu xinu 5.2K May  8 2025 Makefile
-rw-r--r-- 1 xinu xinu  68 Jun  1 03:32 version
-rwxr-xr-x 1 xinu xinu 156K Jun  1 03:32 xinu.elf
-rw-r--r-- 1 xinu xinu 152K Jun  1 03:32 xinu.map
xinu@xinu-develop-end:~/xinu/compile$
```

Dalam gambar tersebut, dapat dilihat bahwa xinu.elf (hasil kompilasi xinu) berukuran 156KB.

2. Membaca source code Xinu



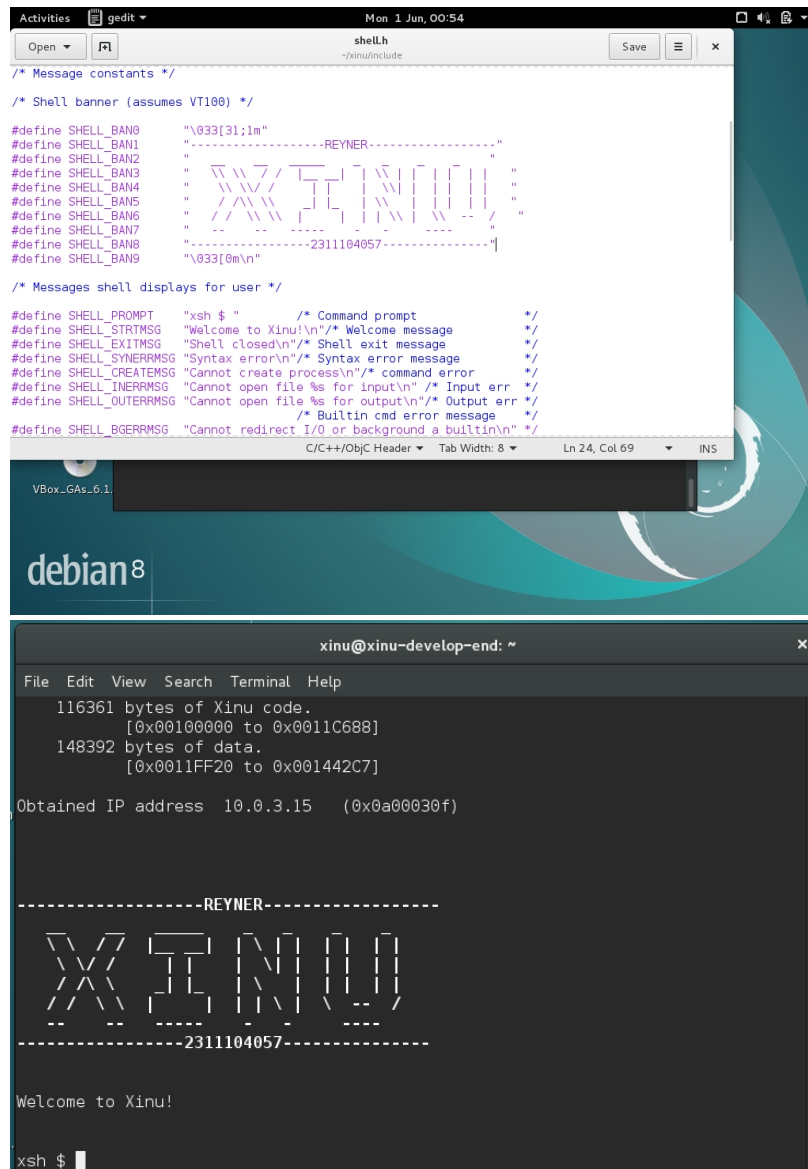
3. Carilah struktur data dari proses pada Xinu OS. Struktur data proses ada pada file apa? Informasi apa saja yang disimpan dalam struktur data tersebut?

```
xinu@xinu-develop-end:~$ grep -r "struct procent" xinu/include/
xinu/include/process.h:struct procent {          /* Entry in the process table */
/
xinu@xinu-develop-end:~$
```

Struktur data tersebut bernama procent (singkatan dari process entry). Di dalamnya tersimpan informasi penting untuk penjadwalan proses, seperti:

- prstate: Status dari proses (misalnya PR_CURR untuk proses yang sedang berjalan, atau PR_READY).
- prprio: Tingkat prioritas proses.
- prstkptr & prstkbase: Informasi pointer dan basis dari stack memori proses tersebut.
- prname: Nama dari proses yang berjalan.

4. Mengubah welcome banner pada Xinu



The image shows a gedit editor window titled 'shell.h' with the following code:

```
/* Message constants */

/* Shell banner (assumes VT100) */

#define SHELL_BAN0 "\033[31;1m"
#define SHELL_BAN1 "-----REYNER-----"
#define SHELL_BAN2 " "
#define SHELL_BAN3 " X X X X X X X X X X X X X X X X "
#define SHELL_BAN4 " X X X X X X X X X X X X X X X X "
#define SHELL_BAN5 " X X X X X X X X X X X X X X X X "
#define SHELL_BAN6 " X X X X X X X X X X X X X X X X "
#define SHELL_BAN7 " X X X X X X X X X X X X X X X X "
#define SHELL_BAN8 "-----2311104057-----"
#define SHELL_BAN9 "\033[0m\n"

/* Messages shell displays for user */

#define SHELL_PROMPT "xsh $" /* Command prompt */
#define SHELL_STRMSG "Welcome to Xinu!\n" /* Welcome message */
#define SHELL_EXITMSG "Shell closed!\n" /* Shell exit message */
#define SHELL_SYNERMSG "Syntax error!\n" /* Syntax error message */
#define SHELL_CREATMSG "Cannot create process!\n" /* command error */
#define SHELL_INERMSG "Cannot open file %s for input!\n" /* Input err */
#define SHELL_OUTERMSG "Cannot open file %s for output!\n" /* Output err */
#define SHELL_BUILTMSG "Built-in cmd error message" /* Builtin cmd error message */
#define SHELL_BGERMSG "Cannot redirect I/O or background a builtin!\n" /*
```

The terminal window shows the output of the Xinu boot process:

```
xinu@xinu-develop-end: ~
File Edit View Search Terminal Help
116361 bytes of Xinu code.
[0x00100000 to 0x0011C688]
148392 bytes of data.
[0x0011FF20 to 0x001442C7]
Obtained IP address 10.0.3.15 (0xa00030f)

-----REYNER-----
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 X X X X X X X X X X X X X X X X
 X X X X X X X X X X X X X X X X
-----2311104057-----

Welcome to Xinu!

xsh $
```